SEOP Technology Stack

Polyglot, best-of-breed, no vendor lock-in — integrated through standards and open APIs

Data Sources

Relational databases (PostgreSQL, MySQL, Oracle, SQL Server, SAP HANA)
Semi-structured: JSON · XML · CSV · Parquet · YAML
Documents: PDF · DOCX · PPTX · HTML · Scanned images (OCR) · Knowledge graphs (RDF, property graphs)

Ingestion & Parsing

LlamaParse · Unstructured.io · Apache Tika · PyMuPDF · pdfplumber · Camelot
Tesseract OCR · Azure Document Intelligence · Debezium (CDC) · Kafka Connect · Airbyte · Singer
JDBC / ODBC adapters · REST / GraphQL connectors · S3 / GCS / ADLS connectors

Processing · NLP · LLM

spaCy · GLiNER · Hugging Face Transformers · sentence-transformers
LLMs: OpenAI GPT-4o · Anthropic Claude 3.5 · Google Gemini 2.5
Self-hosted: Llama 3 · Mistral · Qwen
Embeddings: text-embedding-3-large · BGE-M3 · Cohere Embed v3
Vector search: FAISS · Qdrant · Weaviate · pgvector
Clustering: HDBSCAN · UMAP · scikit-learn

Ontology Engineering & Validation

OWLready2 · RDFLib · Apache Jena · OWL API · ROBOT
Reasoners: HermiT · Pellet · ELK · Openllet
Validation: pyshacl · ShEx.js · OOPS! · OntoClean
Mapping: r2rml-parser · d2rq · morph-kgc (RML) · Direct Mapping
Serialization: Turtle · RDF/XML · OWL Manchester · JSON-LD
Standards: BFO · DOLCE · ISO/IEC 21838 · ISO 29002 · Schema.org · SKOS · PROV-O

Storage · Query · Graph

Triple stores: Neo4j · GraphDB · Stardog · Jena TDB2 · Blazegraph · Virtuoso
SPARQL 1.1 endpoints · GraphQL (Strawberry) · REST / OpenAPI
PostgreSQL (metadata, provenance event log) · ClickHouse (analytics) · DuckDB (ad-hoc)
Object storage: S3 / GCS / ADLS (raw data, model artifacts, scenario library)

Statistical Rigor & Evaluation

SciPy · NumPy · pandas · scikit-posthocs (Nemenyi) · statsmodels
Bootstrap CI (10,000 samples) · Friedman test · critical-difference diagrams (Matplotlib)
BURR benchmark harness (pytest + Hydra) · scenario runner · score matrix exporter
Dual-metric engine (lexical F1 + mapping F1) · divergence detector

Orchestration · APIs · Frontend

Orchestration: LangGraph · LangChain · Prefect · Apache Airflow · Temporal
APIs: FastAPI · gRPC · Pydantic · GraphQL (Strawberry) · Webhooks · Kafka events
Frontend: Next.js · React · Tailwind · Cytoscape.js (graph viz) · Monaco editor (SPARQL)

Security · Identity · Governance

Keycloak · Auth0 · OAuth2 / OIDC · OPA (Open Policy Agent)
HashiCorp Vault (secrets) · Presidio (PII redaction) · Trivy · Falco · SPIFFE
RBAC: row/column/attribute-level · JIT access · MFA · audit logging (OpenTelemetry)
Compliance: SOC 2 · ISO 27001 · GDPR · HIPAA · CMMI-5 evidence auto-collection

Deployment · Ops · Observability

Docker · Kubernetes · Helm · Terraform · ArgoCD · GitHub Actions · GitLab CI · Jenkins
Observability: OpenTelemetry · Prometheus · Grafana · Jaeger · LangSmith · LangFuse
Provenance: PROV-O · Git LFS · DVC · SemVer · PostgreSQL event log · Kafka
MLOps: MLflow · Weights & Biases · DVC · model registry

Standards-first integration: every component connects through W3C standards (RDF, OWL, SPARQL, SHACL, PROV-O, R2RML) and open APIs.

No proprietary lock-in: swap LLM providers, triple stores, or orchestrators without rewriting the pipeline.

All components are production-proven in enterprise knowledge-graph and semantic-platform deployments.